Perception in Bird’s eye view (BEV) using Inverse perspective Mapping (IPM)

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Motivation

Very helpful in teleoperation because of the following reasons:-

Improved Situational Awareness

- Provides a clear **top-down perspective** of the environment.
- Helps drivers or autonomous systems see road edges more easily.

Enhanced Navigation & Parking

- BEV simplifies **maneuvering in tight spaces** (e.g., parallel parking, traffic, reversing).
- Surround view makes it easier to judge distances and angles.

Seamless Multi-Camera Integration

- Combines front, left, and right cameras into a **single stitched image**.
- Eliminates the need to check multiple camera feeds individually.

Robustness with Feature & Fallback Methods

- Works in **feature-rich areas** (roads with texture, markings).
- Still functional in **feature-poor areas** (parking lots, low light).

BEV (Bird's Eye View) is currently state of the art (SOTA) in autonomous driving perception and planning

Standard Representation in AV Research

- Almost every major AV company (Tesla, Waymo, Cruise, Baidu Apollo, NVIDIA) has moved toward BEV-based perception.
- Academic SOTA models like **BEVFormer, BEVDet, PETR, Lift-Splat-Shoot** are all BEV-centric.
- BEV is now the **default benchmark representation** in top self-driving datasets (nuScenes, Waymo Open, Argoverse).

1. Sensor Fusion into BEV

- AVs have multiple cameras, LiDARs, and radars around the car.
- The data from these sensors is projected into a **BEV space** (a flat, top-down grid map).
- This removes perspective distortion and makes it easier to measure distances and positions.

2. Object Detection & Tracking in BEV

- In BEV, the system detects cars, pedestrians, cyclists, road signs, lane markings, etc.
- Tracking objects in BEV is easier since their motion can be represented in consistent world coordinates.

3. Lane & Road Layout Understanding

- Lane lines and drivable areas are detected from camera or LiDAR features, then mapped into BEV.
- This helps the AV know: *where is the road, which lanes exist, where it can drive, and where obstacles are*.
- For example: Tesla's Autopilot and FSD stack heavily rely on **BEV lane reconstruction**.

4. Prediction & Planning in BEV

- Once objects and lanes are in BEV, the system predicts how they will move (e.g., a pedestrian crossing, a car changing lanes).
- The planner then decides safe paths for the AV using this **map-like BEV space**.
- This makes motion planning much more robust than using raw camera images.

5. End-to-End Neural Networks in BEV

- Modern approaches (e.g., Waymo, Tesla, NVIDIA DRIVE) use **BEV-based deep learning models**.
- Cameras → Neural Network → BEV features → Object Detection + Lane + Planning.
- BEVNet, BEVFormer, and LSS (Lift-Splat-Shoot) are popular architectures.

BEV - https://www.youtube.com/watch?v=q9jbiiC9fVo

Approach

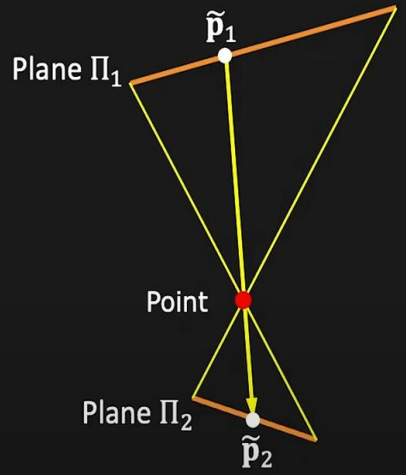
Projective Transformation

Any transformation of the form:

$$\begin{bmatrix}\tilde{x}_2\\ \tilde{y}_2\\ \tilde{z}_2\end{bmatrix} = \begin{bmatrix}h_{11} & h_{12} & h_{13}\\ h_{21} & h_{22} & h_{23}\\ h_{31} & h_{32} & h_{33}\end{bmatrix}\begin{bmatrix}\tilde{x}_1\\ \tilde{y}_1\\ \tilde{z}_1\end{bmatrix} \qquad \tilde{\mathbf{p}}_2 = H\tilde{\mathbf{p}}_1$$

Also called Homography

Mapping of one plane to another through a point

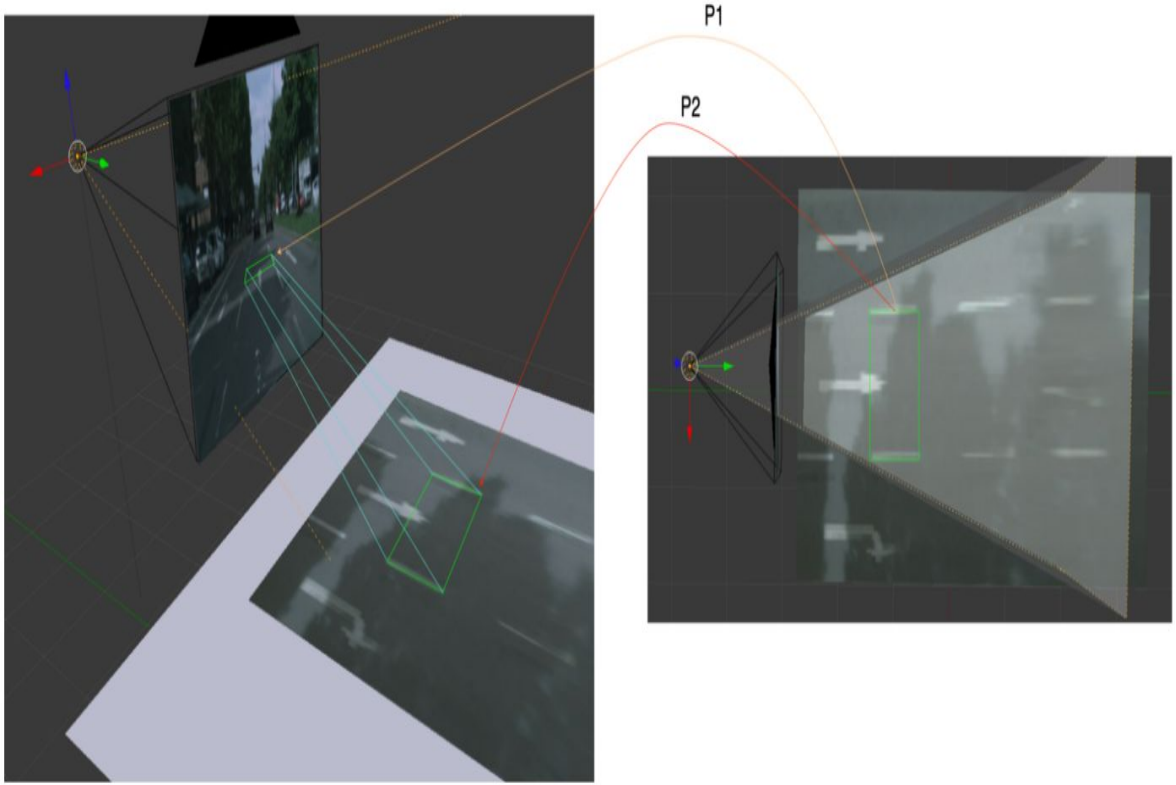


$$\tilde{\mathbf{p}}_2 = H\tilde{\mathbf{p}}_1$$

$$\begin{bmatrix}\tilde{x}_2\\ \tilde{y}_2\\ \tilde{z}_2\end{bmatrix} = \begin{bmatrix}h_{11} & h_{12} & h_{13}\\ h_{21} & h_{22} & h_{23}\\ h_{31} & h_{32} & h_{33}\end{bmatrix}\begin{bmatrix}\tilde{x}_1\\ \tilde{y}_1\\ \tilde{z}_1\end{bmatrix}$$

Homography

Visual representation of IPM




```
src = np.float32([
    [700, 900],  #
    [1300, 900], #
    [100, 1500], #
    [1950, 1500] #
])

dst = np.float32([
    [500, 500],
    [1500, 500],
    [500, 1300],
    [1500, 1300]
])
```

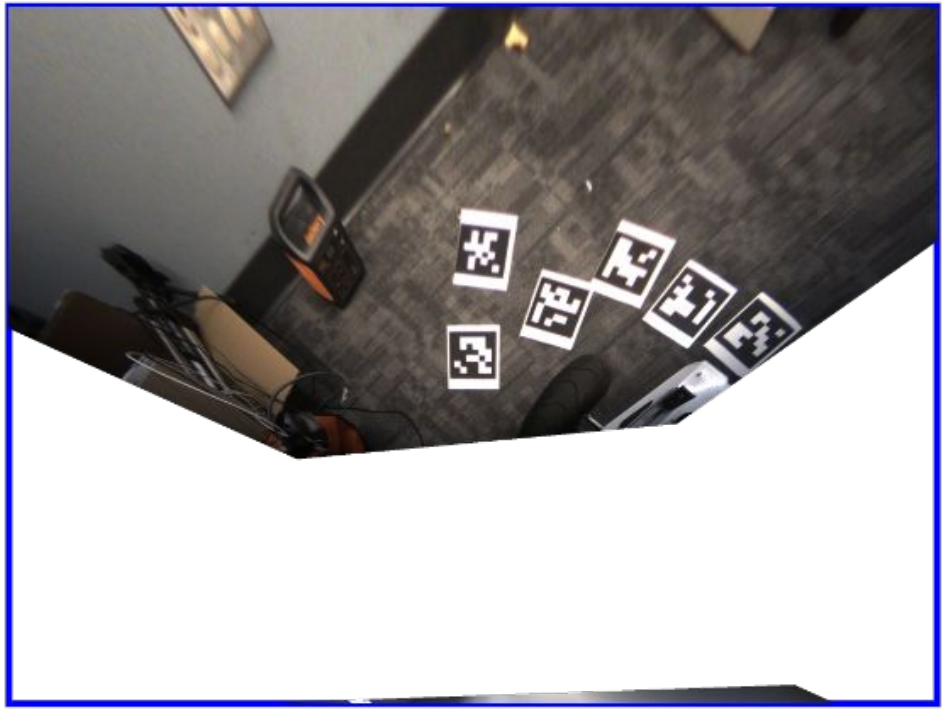
we calculate the M(Transformation matrix) using src and dst and then apply it to image to get BEV

RESULTS

Front normal

Front BEV

Left normal



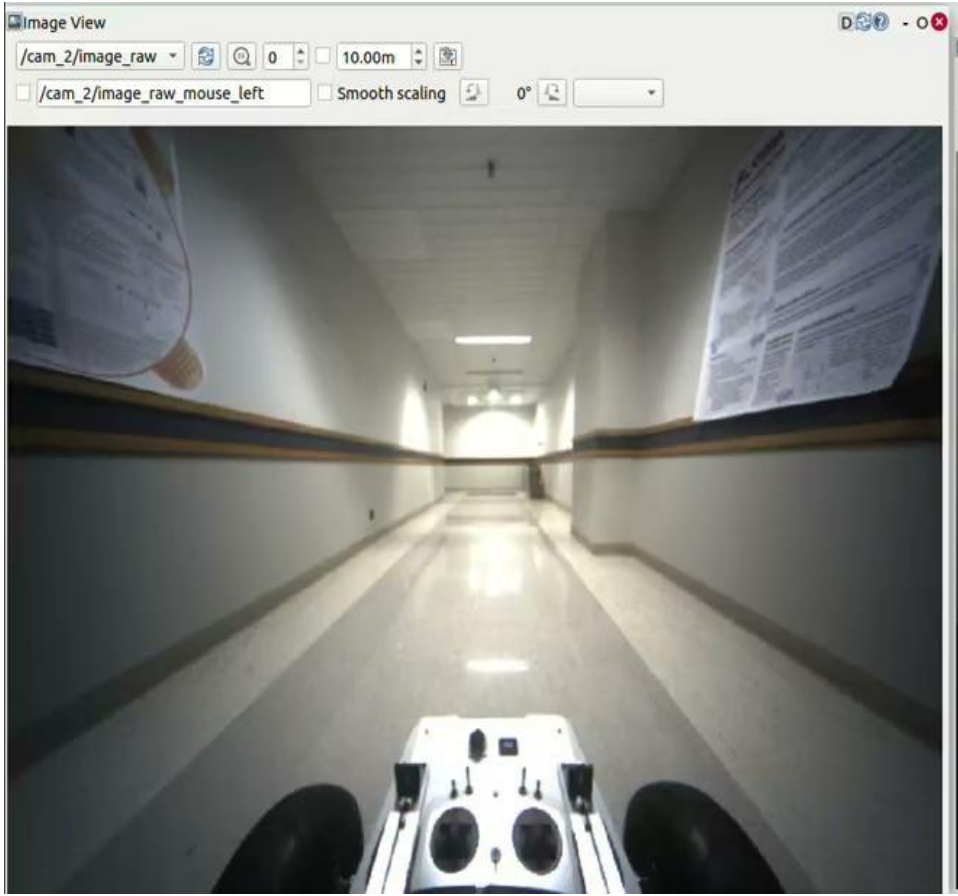
Left BEV

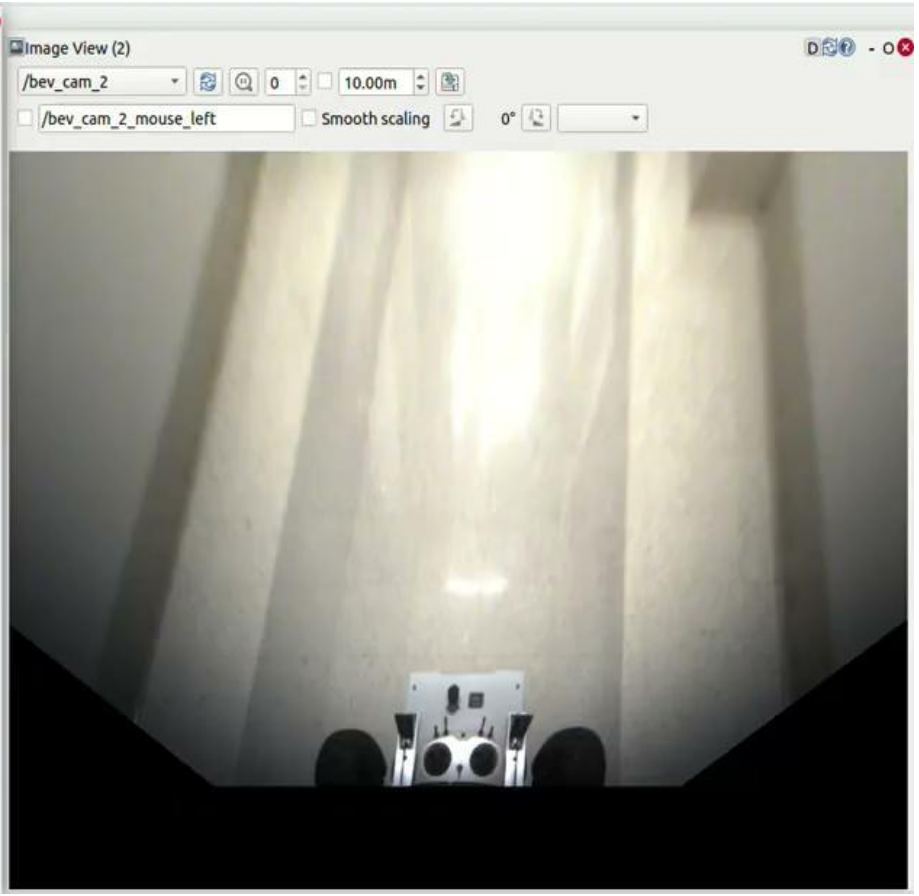
Right normal

Right BEV

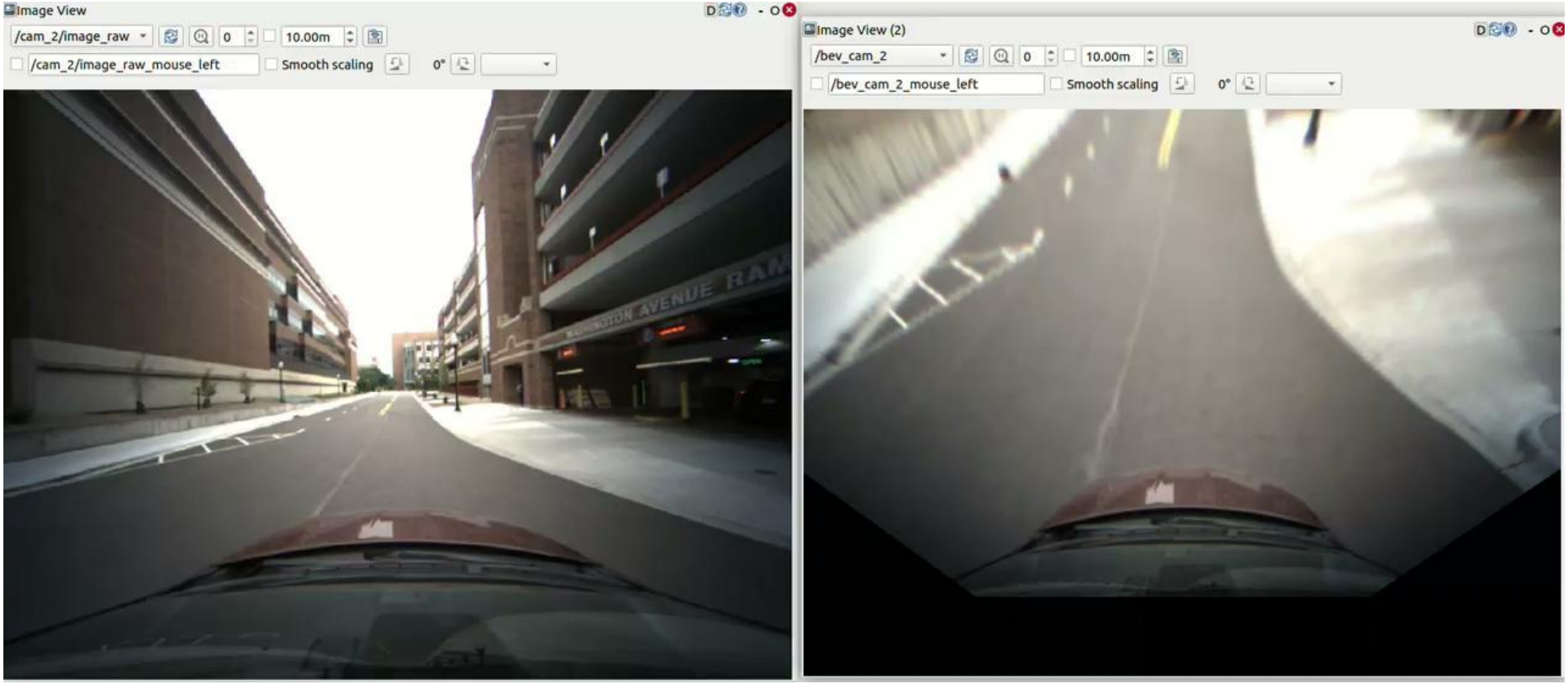
Complete stitched BEV

Real time BEV





BEV on MnCAV VEHICLE (OUTDOOR)



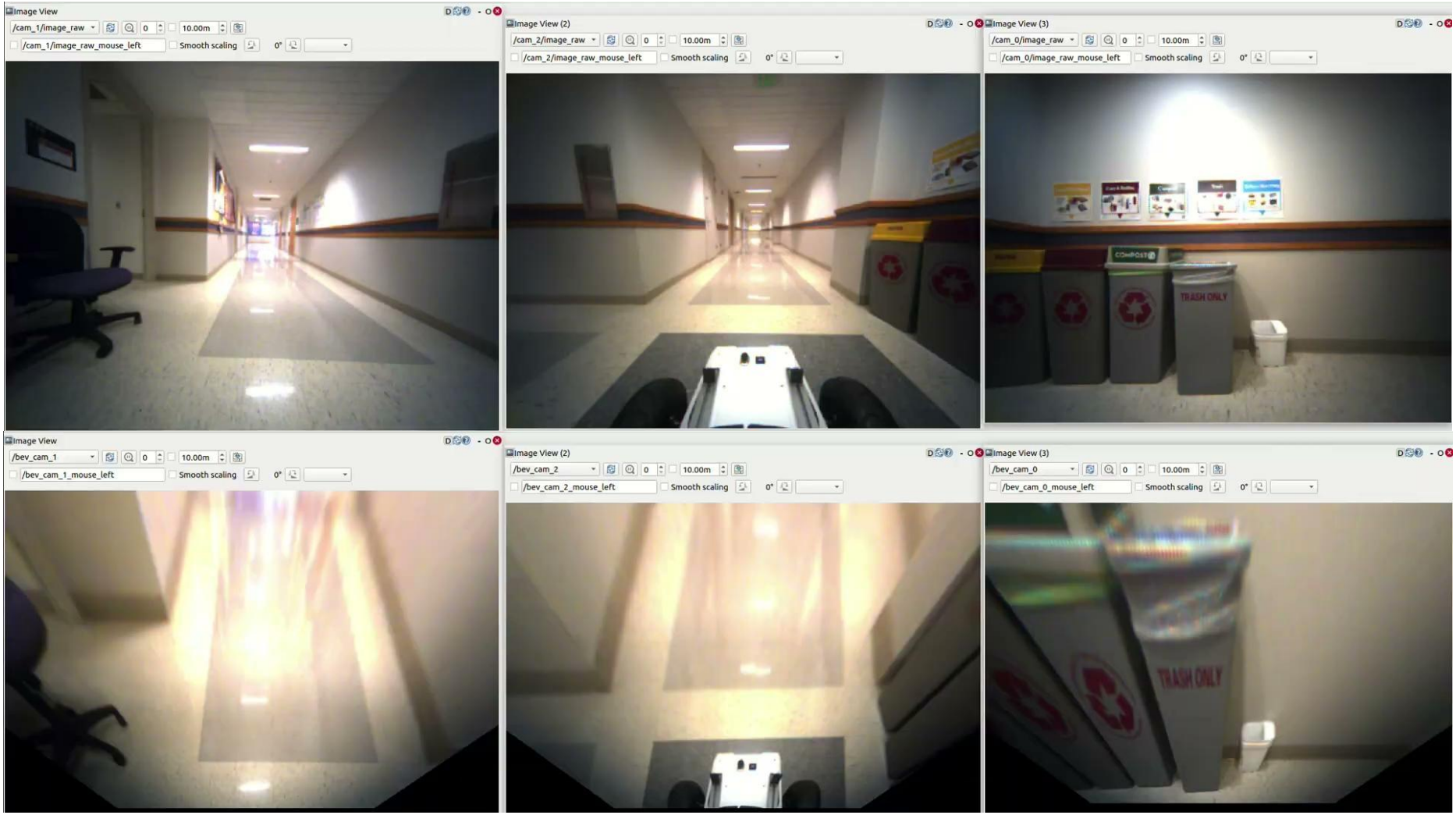


Image View
/cam_1/image_raw
0
10.00m
/cam_1/image_raw_mouse_left
Smooth scaling
0°
Image View (2)
/cam_2/image_raw
0
10.00m
/cam_2/image_raw_mouse_left
Smooth scaling
0°
Image View (3)
/cam_0/image_raw
0
10.00m
/cam_0/image_raw_mouse_left
Smooth scaling
0°
TRASH ONLY
Image View
/bev_cam_1
0
10.00m
/bev_cam_1_mouse_left
Smooth scaling
0°
Image View (2)
/bev_cam_2
0
10.00m
/bev_cam_2_mouse_left
Smooth scaling
0°
Image View (3)
/bev_cam_0
0
10.00m
/bev_cam_0_mouse_left
Smooth scaling
0°
TRASH ONLY

Future Work:

Latency-Aware Rendering

Always time-stamp each camera frame reproject to a unified BEV at the operator side with latency compensation.

Time-Synchronized Projection

Every BEV frame carries sensor timestamps.

Instead of projecting raw latest data, interpolate/extrapolate poses of the ego vehicle and nearby objects to a **common "virtual now"** based on measured latency.

This avoids misalignment of objects across multiple sensors with different delays.

Adaptive Update Rate

If RTT is high, instead of sending full BEV frames, transmit **delta updates** or compressed **vector maps** (bounding boxes, drivable areas).